

ALERT meeting

Time reconstruction of saturated signals

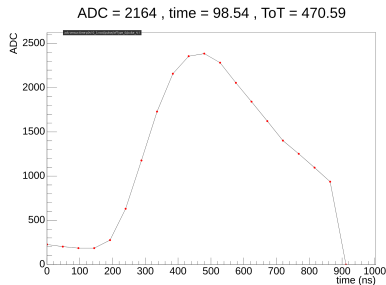
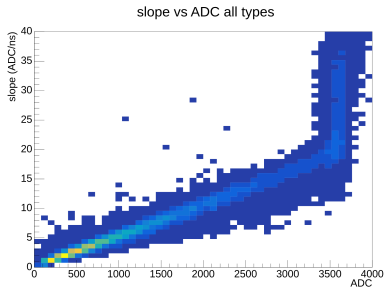
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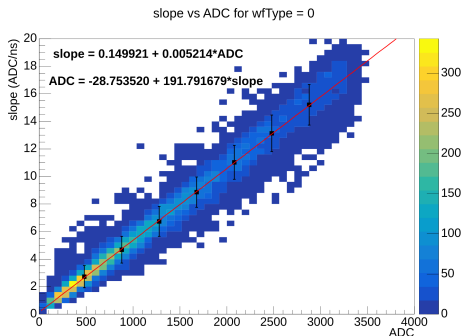
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- ▶ For non-saturated signals, the ADC is proportional the slope (here the maximum of the derivative of the pulse)
- ▶ We can now predict the ADC value of saturated signals

$$ADC_{corr} = f(slope)$$



► Fit result

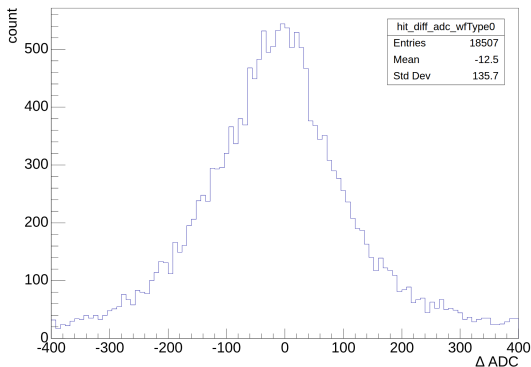


► Next steps

- Recover the Time and ToT from ADC_{corr}
- Implement in coatjava

► Delta ADC for wfType 0

$\Delta \text{ADC} = \text{raw ADC} - \text{predicted ADC}$



- Bias in the time reconstruction of saturated signals
- For such signals, the arrival time and the timeOverThreshold can be wrong

